

FAGOR 8055i CNC

Ordering Handbook.

Ref. 1111



CNC FAGOR 8055i

FAGOR 
FAGOR AUTOMATION

CNC 8055i BASIC DESCRIPTION

The 8055i CNC is a compact CNC (LCD Monitor and CPU behind) available in two versions:

VERSION 1

LCD Monitor + CPU + KEYBOARD + OPERATOR PANEL IN ONE UNIT

KEYBOARD + OPERATOR PANEL

CPU



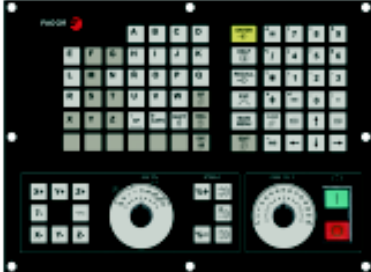
VERSION 2

LCD Monitor + CPU and KEYBOARD + OPERATOR PANEL SEPARATE

CPU



KEYBOARD + OPERATOR PANEL



CNC 8055i: CONFIGURATIONS

**LCD Monitor + CPU + Keyboard +
Operator panel in ONE UNIT**



FL or Power

CNC 8055i Power-TC-K

GP : General propourse
M : Milling machines & Centers
T : Lathes and Turning Centers
MC : Conversational model for Milling
TC : Conversational model for Lathes
EN : Engraving machines

CNC 8055i FL-GP-K	83080122
CNC 8055i FL-M-K	83080124
CNC 8055i FL-MC-K	83080125
CNC 8055i FL-T-K	83080127
CNC 8055i FL-TC-K	83080128
CNC 8055i FL-EN-K	83080093

CNC 8055i Power-GP-K	83080130
CNC 8055i Power-M-K	83080133
CNC 8055i Power-MC-K	83080134
CNC 8055i Power-T-K	83080136
CNC 8055i Power-TC-K	83080137

**LCD Monitor + CPU + Keyboard +
Operator panel SEPARATE**



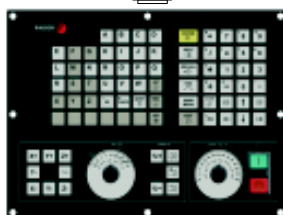
LCD Monitor + CPU

CNC 8055i FL-GP	83080121
CNC 8055i FL-M	83080123
CNC 8055i FL-T	83080126

CNC 8055i Power-GP	83080129
CNC 8055i Power-M	83080132
CNC 8055i Power-T	83080135



0,8 m long cable is included in the basic configuration
Optional 2m long cable **C.JOG 40/55i 2M 03630100**



Keyboard + Operator panel

OP-8055-ALFA	83540044	For Milling machines, Centers, Lathes and Turning Centers
OP-8055-MC	83540045	For Milling Conversational models
OP-8055-TC	83540046	For Lathes Conversational models
OP-8055-MCO/TCO	83540047	For Customizable models

MAKING UP THE CENTRAL UNIT

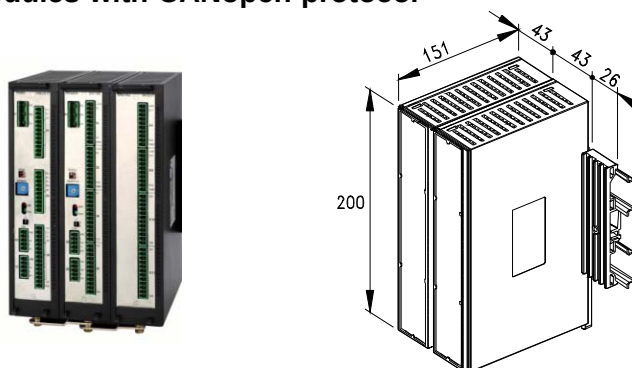
Central Unit	CNC 8055i -x - x - x - x
RS-232	
Digital 16I / 8O's	
2 Handweel inputs	
1 Analog Output & 1 Feedback input for Spindle, Axis or Handweel	
USB as memory expansion for part-programs and PLC programs	

Analog & Local I/O's boards	
Analog board	ANALOG 8055i-C
4 Analog outputs & 4 Feedback inputs for the Axes (Spindles or Handweels)	83070030
Local I/O's board	4OI/24O-8055i-B
Digital 40I/24O's	83400005
Analog & Local I/O's board	ANALOG+40I/24O-C
4 Analog outputs & 4 Feedback inputs for the Axes (Spindles or Handweels)	83070031
Digital 40I/24O's	
Cover	COVER ANA+I/O-8055i-B
When you don't use Analog or Local I/Os boards	83402001

Expansion boards	
ETHERNET-CAN-SERCOS	ETHERNET-CAN-SERCOS
Ethernet communication: Ethenrnet board	83240176
CAN for Remote I/O's : CAN interface for Remote I/O's	
Sercos: Sercos interface for drive system	
ETHERNET-CAN-CANAXES	ETHERNET-CAN-CANAXES
Ethernet communication: Ethenrnet board	83240175
CAN for Remote I/O's :CAN interface for Remote I/O's	
CAN: CAN interface for drive system	
ETHERNET-CAN AXES	ETHERNET-CAN AXES
Ethernet communication: Ethenrnet board	83240177
CAN: CAN interface for drive system	
ETHERNET-BOARD	ETHERNET-BOARD
Ethernet communication: Ethenrnet board	83240169

REMOTE I/O MODULES

RIO5 series. Remote modules with CANopen protocol



Power Supply & 24I/16O	RIO5-PS-24I/16O	83070015
Power Supply & 2 PT100 & 4Ai/4AO	RIO5-PS-2PT100/4AI/4AO	83070018
Expansion module 24I/16O	RIO5-24I/16O EXP MOD	83070016
Expansion module 48I/32O	RIO5-48I/32O EXP MOD	83070017



RIO5-PS-24I/16O 83070015

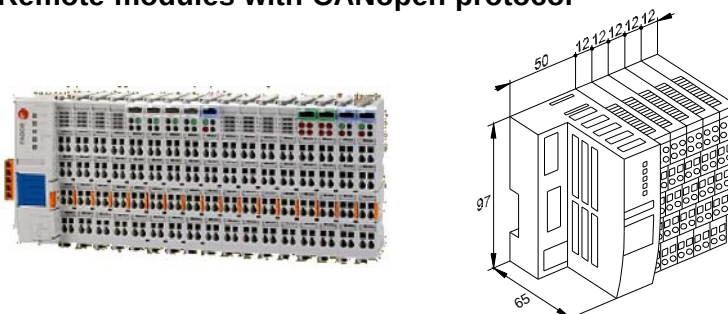


RIO5-PS-24I/16O 83070015
RIO5-24I/16O EXP MOD 83070016



RIO5-PS-24I/16O 83070015
RIO5-48I/32O EXP MOD 83070017

RIOW series. Remote modules with CANopen protocol



Header CANopen	RIOW CANOPEN STAND	03300010
Expansion module with 8 digital inputs	RIOW 8DI-51205053	03300008
Expansion module with 8 digital outputs	RIOW 8DO-51205043	03300007
Expansion module with 4 analog inputs	RIOW 4AI-51205050	03300006
Expansion module with 4 analog outputs	RIOW 4AO-51205039	03300005
Expansion module with 2 PT100 inputs	RIOW 2AI PT100-51205048	03300004
24V _{DC} 10A power supply for the modules	RIOW PS24-51205036	03300003
Line terminating module.	RIOW END-51205037	03300000

CAN cable for Remote I/O's

5m	CAN CABLE 5M	03780100
10m	CAN CABLE 10M	03780101
15m	CAN CABLE 15M	03780102
20m	CAN CABLE 20M	03780103
25m	CAN CABLE 25M	03780104
30m	CAN CABLE 30M	03780106
35m	CAN CABLE 35M	03780107
40m	CAN CABLE 40M	03780108
45m	CAN CABLE 45M	03780109
50m	CAN CABLE 50M	03780110
75m	CAN CABLE 75M	03780111
100m	CAN CABLE 100M	03780112
150m	CAN CABLE 150M	03780113

SERCOS cables

Sercos optic fiber

1m	SFO-1	03900001
2m	SFO-2	03900002
3m	SFO-3	03900003
5m	SFO-5	03900005
7m	SFO-7	03900007
10m	SFO-10	03900010
12m	SFO-12	03900012

Flexible Sercos optic fiber

10m	SFO-FLEX-10	03910010
15m	SFO-FLEX-15	03910015
20m	SFO-FLEX-20	03910020
25m	SFO-FLEX-25	03910025
30m	SFO-FLEX-30	03910030
35m	SFO-FLEX-35	03910035
40m	SFO-FLEX-40	03910040
40m	SFO-V-FLEX-40	03920040
50m	SFO-V-FLEX-50	03920050
60m	SFO-V-FLEX-60	03920060
75m	SFO-V-FLEX-75	03920075
100m	SFO-V-FLEX-100	03920100

Educational CNC

CNC 8055i EDUC-K

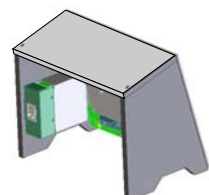
Educational CNC 83080092



Accessories

8055i CNC SUPPORT

Tabletop CNC stand kit 03550000



KEYCF AND SOFTWARE FEATURES.

The KEYCF include the software features by means of a special code.

It must be used to store critical data used for customizing the CNC for that machine, parameters, PLC programs and screens.

The remaining memory space is free for storing User programs.

Basic 512 MB KEYCF

For 8055i FL model **KEY CF 8055 FL BASIC** **83240130**

For 8055i Power model **KEY CF 8055 Power BASIC** **83240132**

Optional 2GB KEYCF

For 8055i FL model **KEY CF 8055 FL LARGE** **83240131**

For 8055i Power model **KEY CF 8055 Power LARGE** **83240133**

Software options for CNC 8055i FL model

Code	Software features	GP	M	MC	MCO	T	TC	TCO	EN
	Block processing time	3.5 ms							
	Loop time	4 ms							
	Look ahead	100 blocks							
80501010	Number of axes with standard software	(4)	(4)	(4)	(4)	(2)	(2)	(2)	(3)
80501000	Number of axes with optional software	(-)	(-)	(-)	(-)	(4)	(4)	(4)	(-)
	Save, Edit & Execute programs in KEYCF	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
	Electronic threading	(-)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
	Tool magazine management	(-)	(1)	(1)	(1)	(1)	(1)	(1)	(-)
	Machining canned cycle	(-)	(1)	(1)	(-)	(1)	(1)	(-)	(1)
	Multiple machining canned cycle	(-)	(1)	(1)	(-)	(-)	(-)	(-)	(1)
83240166	Remote Hard disk	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(1)
83240167	Ethernet	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(1)
80501004	Rigid tapping	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(1)
80501006	Tool life monitoring	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(1)
80501007	Probing canned cycles	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(1)
83240501	Telediagnosis	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(1)
83270003	Tool radius compensation	(-)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
83240145	Tangential Control	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(-)
83240155	Retracing	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(1)
83240220	GP Basic Software	(1)	(-)	(-)	(-)	(-)	(-)	(-)	(-)
83240149	M Basic Software	(-)	(1)	(1)	(1)	(-)	(-)	(-)	(-)
83240141	MC Conversational Software	(-)	(-)	(1)	(-)	(-)	(-)	(-)	(-)
83240143	MCO Conversational Software	(-)	(-)	(-)	(1)	(-)	(-)	(-)	(-)
83240148	T Basic Software	(-)	(-)	(-)	(-)	(1)	(1)	(1)	(-)
83240142	TC Conversational Software	(-)	(-)	(-)	(-)	(-)	(1)	(-)	(-)
83240144	TCO Conversational Software	(-)	(-)	(-)	(-)	(-)	(-)	(1)	(-)
83240220	NE Basic Software	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(1)
80501002	Line and Solid graphics	(-)	(1)	(1)	(1)	(1)	(1)	(1)	(-)
80501003	Pockets with islands	(-)	(1)	(1)	(1)	(-)	(-)	(-)	(-)
80501008	DNC Communications	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
83270002	Profile editor	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(-)
83240156	Setup assistance	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)

(-) Not available

(1) Standard feature

(-) Optional feature

Software options for CNC 8055i Power model

Code	Software features	GP	M	MC	MCO	T	TC	TCO
	Block processing time	1 ms						
	Loop time	2 ms						
	Look ahead	200 blocks						
80501010	Number of axes with standard software	(4)	(4)	(4)	(4)	(2)	(2)	(2)
80501000	Number of axes with optional software	(7)	(7)	(7)	(7)	(4)	(4)	(4)
80501001	Number of axes with optional software	(-)	(-)	(-)	(-)	(7)	(7)	(7)
	Save, Edit & Execute programs in KEYCF	(1)	(1)	(1)	(1)	(1)	(1)	(1)
	Electronic threading	(-)	(1)	(1)	(1)	(1)	(1)	(1)
	Tool magazine management	(-)	(1)	(1)	(1)	(1)	(1)	(1)
	Machining canned cycle	(-)	(1)	(1)	(-)	(1)	(1)	(-)
	Multiple machining canned cycle	(-)	(1)	(1)	(-)	(-)	(-)	(-)
83240166	Remote Hard disk	(-)	(-)	(-)	(-)	(-)	(-)	(-)
83240167	Ethernet	(-)	(-)	(-)	(-)	(-)	(-)	(-)
80501004	Rigid tapping	(-)	(-)	(-)	(-)	(-)	(-)	(-)
80501005	Digitizing	(-)	(-)	(-)	(-)	(-)	(-)	(-)
80501006	Tool life monitoring	(-)	(-)	(-)	(-)	(-)	(-)	(-)
80501007	Probing canned cycles	(-)	(-)	(-)	(-)	(-)	(-)	(-)
80501009	4-axis interpolation	(-)	(-)	(-)	(-)	(-)	(-)	(-)
83240501	Telediagnosis	(-)	(-)	(-)	(-)	(-)	(-)	(-)
83270001	C Axis for Lathe	(-)	(-)	(-)	(-)	(-)	(-)	(-)
83270003	Tool radius compensation	(-)	(1)	(1)	(1)	(1)	(1)	(1)
83270009	Y Axis for Lathe	(-)	(-)	(-)	(-)	(-)	(-)	(-)
83240500	Opción TCP // TCP Option	(-)	(-)	(-)	(-)	(-)	(-)	(-)
83240145	Tangential Control	(-)	(-)	(-)	(-)	(-)	(-)	(-)
83240155	Retracing	(-)	(-)	(-)	(-)	(-)	(-)	(-)
83240220	GP Basic Software	(1)	(-)	(-)	(-)	(-)	(-)	(-)
83240149	M Basic Software	(-)	(1)	(1)	(1)	(-)	(-)	(-)
83240141	MC Conversational Software	(-)	(-)	(1)	(-)	(-)	(-)	(-)
83240143	MCO Conversational Software	(-)	(-)	(-)	(1)	(-)	(-)	(-)
83240148	T Basic Software	(-)	(-)	(-)	(-)	(1)	(1)	(1)
83240142	TC Conversational Software	(-)	(-)	(-)	(-)	(-)	(1)	(-)
83240144	TCO Conversational Software	(-)	(-)	(-)	(-)	(-)	(-)	(1)
80501002	Line and Solid graphics	(-)	(1)	(1)	(1)	(1)	(1)	(1)
80501003	Pockets with islands	(-)	(1)	(1)	(1)	(-)	(-)	(-)
80501008	DNC Communications	(1)	(1)	(1)	(1)	(1)	(1)	(1)
83270002	Profile editor	(1)	(1)	(1)	(1)	(1)	(1)	(1)
83240156	Setup assistance	(1)	(1)	(1)	(1)	(1)	(1)	(1)

(-) Not available

(1) Standard feature

() Optional feature

IN ORDER TO COMPLETE ITS ORDER

ONE UNIT: LCD Monitor + CPU + Keyboard + Operator panel

83080122	CNC 8055i FL-GP-K
83080124	CNC 8055i FL-M-K
83080125	CNC 8055i FL-MC-K
83080127	CNC 8055i FL-T-K
83080128	CNC 8055i FL-TC-K

83080130	CNC 8055i Power-GP-K
83080133	CNC 8055i Power-M-K
83080134	CNC 8055i Power-MC-K
83080136	CNC 8055i Power-T-K
83080137	CNC 8055i Power-TC-K

SEPARATE: LCD Monitor + CPU

83080121	CNC 8055i FL-GP
83080123	CNC 8055i FL-M
83080126	CNC 8055i FL-T

83080129	CNC 8055i Power-GP
83080132	CNC 8055i Power-M
83080135	CNC 8055i Power-T

Keyboard + Operator panel

83540044	OP-8055-ALFA
83540045	OP-8055-MC
83540046	OP-8055-TC
83540047	OP-8055-MCO/TCO

Analog & Local I/O's boards

83070030	ANALOG 8055i-C	4 Analog outputs & 4 Feedback inputs
83400005	40I/24O-8055i-B	40I / 24O
83070031	ANALOG+40I/24O-C	4 Analog outputs, 4 Feedback inputs, 40I / 24O
83402001	COVER ANA+I/O-8055i-B	Cover

Expansion boards

83240176	ETHERNET-CAN-SERCOS
83240175	ETHERNET-CAN-CAN AXES
83240177	ETHERNET-CAN-AXES
83240169	ETHERNET-BOARD

KEYCF

83240130	KEY CF 8055 FL BASIC	83240131	KEY CF 8055 FL LARGE
83240132	KEY CF 8055 Power BASIC	83240133	KEY CF 8055 Power LARGE

SOFTWARE FEATURES

80501010	4 axes for Milling / 2 axes for Lathe	
80501000	7 axes for Milling / 4 axes for Lathe	7 axes only for CNC 8055i Power
80501001	7 axes for Lathe	Only for CNC 8055i Power
83240166	Remote Hard disk	
83240167	Ethernet	
80501004	Rigid tapping	
80501005	Digitizing	Only for M, M, MCO models in CNC 8055i Power
80501006	Tool life monitoring	No available for GP model
80501007	Probing canned cycles	No available for GP model
80501009	4-axis interpolation limit	Only for CNC 8055i Power
83240501	Telediagnosis	
83270001	C Axis for Lathe	Only for CNC 8055i Power
83270003	Tool radius compensation	
83270009	Y Axis for Lathe	Only for CNC 8055i Power
83240500	TCP Option for Milling	Only for CNC 8055i Power
83240145	Tangential Control	
83240155	Retracing	No available for CNC 8055i FL GP model
83240220	GP Basic Software	
83240149	M Basic Software	
83240141	MC Conversational Software	
83240143	MCO Conversational Software	
83240148	T Basic Software	
83240142	TC Conversational Software	
83240144	TCO Conversational Software	

Educational CNC			
	83080092	CNC 8055i EDUC-K	
ACCESORIES			
	03550000	8055i CNC SUPPORT	Tabletop CNC stand kit
Manuals			
	03M50000	M.CNC 8055 M OEM-CAS	8055 M CNC OEM Manual in Spanish
	03M50001	M.CNC 8055 M OEM-ING	8055 M CNC OEM Manual in English
	03M50004	M.CNC 8055 M OEM-ITA	8055 M CNC OEM Manual in Italian
	03M50010	M.CNC 8055 M USER-CAS	8055 M CNC user Manual in Spanish
	03M50011	M.CNC 8055 M USER-ING	8055 M CNC user Manual in English
	03M50012	M.CNC 8055 M USER-FRA	8055 M CNC user Manual in French
	03M50013	M.CNC 8055 M USER-ALE	8055 M CNC user Manual in German
	03M50014	M.CNC 8055 M USER-ITA	8055 M CNC user Manual in Italian
	03M50015	M.CNC 8055 M USER-BRA	8055 M CNC user Manual in Portuguese-Brazilian
	03M50016	M.CNC 8055 M USER-HOL	8055 M CNC user Manual in Dutch
	03M50020	M.CNC 8055 T OEM-CAS	8055 T CNC OEM Manual in Spanish
	03M50021	M.CNC 8055 T OEM-ING	8055 T CNC OEM Manual in English
	03M50024	M.CNC 8055 T OEM-ITA	8055 T CNC OEM Manual in Italian
	03M50030	M.CNC 8055 T USER-CAS	8055 T CNC user Manual in Spanish
	03M50031	M.CNC 8055 T USER-ING	8055 T CNC user Manual in English
	03M50032	M.CNC 8055 T USER-FRA	8055 T CNC user Manual in French
	03M50033	M.CNC 8055 T USER-ALE	8055 T CNC user Manual in German
	03M50034	M.CNC 8055 T USER-ITA	8055 T CNC user Manual in Italian
	03M50035	M.CNC 8055 T USER-BRA	8055 T CNC user Manual in Portuguese-Brazilian
	03M50036	M.CNC 8055 T USER-HOL	8055 T CNC user Manual in Dutch
	03M50040	M.CNC 8055 MC USER-CAS	8055 MC CNC user Manual in Spanish
	03M50041	M.CNC 8055 MC USER-ING	8055 MC CNC user Manual in English
	03M50042	M.CNC 8055 MC USER-FRA	8055 MC CNC user Manual in French
	03M50043	M.CNC 8055 MC USER-ALE	8055 MC CNC user Manual in German
	03M50044	M.CNC 8055 MC USER-ITA	8055 MC CNC user Manual in Italian
	03M50045	M.CNC 8055 MC USER-BRA	8055 MC CNC user Manual in Portuguese-Brazilian
	03M50046	M.CNC 8055 MC USER-HOL	8055 MC CNC user Manual in Dutch
	03M50050	M.CNC 8055 TC USER-CAS	8055 TC CNC user Manual in Spanish
	03M50051	M.CNC 8055 TC USER-ING	8055 TC CNC user Manual in English
	03M50052	M.CNC 8055 TC USER-FRA	8055 TC CNC user Manual in French
	03M50053	M.CNC 8055 TC USER-ALE	8055 TC CNC user Manual in German
	03M50054	M.CNC 8055 TC USER-ITA	8055 TC CNC user Manual in Italian
	03M50055	M.CNC 8055 TC USER-BRA	8055 TC CNC user Manual in Portuguese-Brazilian
	03M50056	M.CNC 8055 TC USER-HOL	8055 TC CNC user Manual in Dutch
	03M50060	M.CNC 8055 MCO/TCO USER-CAS	8055 MCO/TCO CNC user Manual in Spanish
	03M50061	M.CNC 8055 MCO/TCO USER-ING	8055 MCO/TCO CNC user Manual in English
	03M50062	M.CNC 8055 MCO/TCO USER-FRA	8055 MCO/TCO CNC user Manual in Spanish
	03M50063	M.CNC 8055 MCO/TCO USER-ALE	8055 MCO/TCO CNC user Manual in Spanish
	03M50064	M.CNC 8055 MCO/TCO USER-ITA	8055 MCO/TCO CNC user Manual in Italian
	03M50065	M.CNC 8055 MCO/TCO USER-BRA	8055 MCO/TCO CNC user Manual in Spanish
	03M50066	M.CNC 8055 MCO/TCO USER-HOL	8055 MCO/TCO CNC user Manual in Spanish
Spare parts			
	83070032	ANALOG 8055i-C (RF)	83240134 KEY CF 8055 FL BASIC (R)
	83402007	40I/24O-8040/55i-B (RF)	83240135 KEY CF 8055 FL LARGE (R)
	83070033	ANALOG+40I/24O-C (RF)	83240136 KEY CF 8055 Power BASIC (R)
	83402002	COVER ANA+I/O-55i-B (R)	83240137 KEY CF 8055 Power LARGE (R)
	83240179	ETH-CAN-SERCOS (R)	
	83240178	ETH-CAN-CAN AXES (R)	

REMOTE I/O MODULES

RIO5 series

83070015	RIO5 PS-24I/16O	Power Supply + 24I/16O
83070018	RIO5 PS-2PT100/4AI/4AO	PS, 2 PT100, 4 analog inputs and 4 analog outputs
83070016	RIO5 24I/16O EXP MOD	24I/16O
83070017	RIO5 48I/32O EXP MOD	48I/32O

RIOW series

03300010	RIOW CANOPEN STAND	Header module CANopen
03300008	RIOW 8DI-51205053	Expansion module with 8 digital inputs
03300007	RIOW 8DO-51205043	Expansion module with 8 digital outputs
03300006	RIOW 4AI-51205050	Expansion module with 4 analog inputs
03300005	RIOW 4AO-51205039	Expansion module with 4 analog outputs
03300004	RIOW 2AI PT100-51205048	Expansion module with 2 inputs for PT100
03300003	RIOW PS24-51205036	24 V DC 10 A power supply for the modules
03300000	RIOW END-51205037	Line terminating module.

CAN cable for Remote I/O's

03780100	CAN CABLE 5M	03780108	CAN CABLE 40M
03780101	CAN CABLE 10M	03780109	CAN CABLE 45M
03780102	CAN CABLE 15M	03780110	CAN CABLE 50M
03780103	CAN CABLE 20M	03780111	CAN CABLE 75M
03780104	CAN CABLE 25M	03780112	CAN CABLE 100M
03780106	CAN CABLE 30M	03780113	CAN CABLE 150M
03780107	CAN CABLE 35M		

Sercos optic fiber cables

03900001	SFO-1	03900007	SFO-7
03900002	SFO-2	03900010	SFO-10
03900003	SFO-3	03900012	SFO-12
03900005	SFO-5		

Flexible Sercos optic fiber cables

03910010	SFO-FLEX-10	03920040	SFO-V-FLEX-40
03910015	SFO-FLEX-15	03920050	SFO-V-FLEX-50
03910020	SFO-FLEX-20	03920060	SFO-V-FLEX-60
03910025	SFO-FLEX-25	03920075	SFO-V-FLEX-75
03910030	SFO-FLEX-30	03920100	SFO-V-FLEX-100
03910035	SFO-FLEX-35		
03910040	SFO-FLEX-40		